DATA ANALYTICS & BUSINESS INTELLIGENCE CAPSTONE

Executive Decision Report:
Retail Margin Turnaround Strategy

Three Sequenced Strategic Decisions to Lift Annual Profit by +$103k and Deliver $200.7k Net Cash Benefit

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AUDIENCE & GOVERNANCE

Executive Leadership Team

Chief Executive Officer & CFO

VP Sales, VP Ops, VP E-Commerce

Reporting Period: October 2026

DATASET & AUDIT STATUS

11,996 Orders (2023–2025)

Scope: Omnichannel Retail ($3.37M Sales)

Status: Audited Stage 4 Financial Engine

Consistency Audit Status: PASSED (75/75)

EXECUTIVE SUMMARY

Executive Decision Brief: Reversing Margin Decay

Revenue expanded +56% since 2023, but profit margin collapsed from 15.4% to 11.9%. Three sequenced turnaround decisions restore financial health.

1. THE BUSINESS PROBLEM

Growth Is Unprofitable

- Revenue climbed +56.2% ($897k -> $1.40M), but contribution profit rose only +20.7%.
- Weighted margin collapsed -3.51 pp from 15.44% in 2023 down to 11.93% in 2025.

Root Cause 1: Pricing Indiscipline

- Discounts >20% generate -$40,094 in direct annual losses across 1,605 transactions.

Root Cause 2: Marketplace Fee Drag

- Marketplace sales surged to 26% share but yields -2.55% negative margin after 15% fees.

Root Cause 3: Shipping Delay Crisis

- Central region ships 58.4% of orders late, tripling returns (14.9%) and cutting retention.

2. THE THREE STRATEGIC DECISIONS

Decision 1: Automated Discount Guardrails

- Hard cap discounts at 20%; 15% basket floor.
- Uplift: +$66,332/yr | Capex: $10k | M3 Payback.

Decision 2: Marketplace Margin Repair

- Renegotiate fee to 14% (-1 pp); cap discount 10%.
- Uplift: +$29,352/yr | Capex: $12k | M12 Payback.

Decision 3: Delivery SLA & Retention

- Central 3PL overhaul; reduce late to 15%.
- Uplift: +$7,314/yr | Capex: $12k | Strategic SLA.

Sequenced Execution Logic:

- Month 1–3: Rec 1 delivers fast cash surplus.
- Month 4–6: Rec 2 fixes third-party bleed.
- Month 7–12: Rec 3 stabilizes brand loyalty.

Zero Double-Counting Principle:

- Rec 1: Pricing governance on non-MP orders.
- Rec 2: MP fee reduction on non-discounted.
- Rec 3: Return reduction on late deliveries only.

3. EXPECTED VALUE & DISCLOSURES

Annual Profit Uplift (Steady-State)	**+$102,997/yr**
36-Month Net Cash Benefit	**+$200,723**
Net Return on Investment (ROI)	**232.72%**
Discounted NPV (@ 10% WACC)	**+$168,399**
Portfolio Capital Payback	**Month 8**

CRITICAL EXECUTIVE DISCLOSURES:

1. Rec 3 Standalone Deficit: -$18,795 over 36m. Opex ($9.0k/yr) exceeds benefit ($7.3k/yr). Justified as strategic SLA investment funded by Rec 1 ($171k) & Rec 2 ($48k) cash surplus.
2. Rec 1 Contagion Risk: 1,605 loss orders belong to 825 buyers who ALSO generated $741k in profitable sales. Use basket margin rules to prevent customer churn contagion.

COMMERCIAL SETTING

Business Context & Strategic Objectives

Analyzing 11,996 transactions across three fulfillment channels, five geographic territories, and three product categories.

Omnichannel Retail Architecture

Operational mix across 3 commercial channels:

Direct Retail Store

3,869 Orders | $1.16M Sales (34.4%)

Profit: $245.9k | Weighted Margin: 21.22%

Strongest margin profile; physical stores act as high-margin anchors with strong brand loyalty.

Online Direct (Web)

5,467 Orders | $1.53M Sales (45.3%)

Profit: $229.6k | Weighted Margin: 15.00%

Primary volume driver; stable direct-to-consumer digital channel with solid unit profitability.

Third-Party Marketplace

2,660 Orders | $684.1k Sales (20.3%)

Profit: -$17.4k | Weighted Margin: -2.55%

Fastest growing channel (+14 pp share) but losing money due to 15% platform fees & discounting.

Core Business Questions

Strategic investigation guiding executive action:

Q1: Revenue vs Profit Decoupling

Why has rapid top-line growth (+56% over 3 years) produced margin compression (-3.5 pp)? What operational mechanisms caused dollar profit to lag volume?

Q2: Loss Concentration & Subsidies

Which discount bands, product categories, and sales channels are destroying capital? Where does cross-subsidization hide critical business risks?

Q3: Delivery Quality & Customer Attrition

Does shipping latency correlate with merchandise returns and customer churn? How much margin is lost through supply chain failure?

Q4: Actionable Turnaround Roadmap

What concrete, sequenced interventions can management execute within 12 months with audited positive ROI and minimal customer churn?

Strategic Imperatives & Scope

Guiding principles for the capstone engagement:

Shift from Volume to Profit

Replace gross revenue targets with net contribution profit after channel fees, freight, and returns.

Stop Customer-Level Contagion

Do not bluntly cancel orders; apply basket-level minimum margin rules to preserve profitable relationships.

Hold Third-Party Partners Accountable

Renegotiate marketplace commission fees and 3PL carrier SLA contracts backed by penalty clauses.

Protect Repeat Customer Trust

Repeat buyers generate 95% of orders. Service delivery quality is the primary driver of customer lifetime value.

GOVERNANCE & PIPELINE

Data Governance, Cleaning Pipeline & Analytics Methodology

A reproducible, 6-stage deterministic pipeline transforming 12,400 raw entries into 11,996 audited, reliable transaction records.

Dataset Coverage & Scope

Operational parameters of the analysis:

Reporting Window

3 Calendar Years (Jan 1, 2023 – Dec 31, 2025)

Raw Transaction Volume

12,400 initial transaction rows across 20 fields

Clean Transaction Volume

11,996 fully reconciled rows (zero missing cells)

Geographic Scope

5 US Regions: West, East, North, South, Central

Merchandise Scope

3 Categories: Furniture, Office Supplies, Technology

Customer Base

2,116 unique customers across B2C & B2B segments

SIMULATED-DATA DISCLOSURE:

- Dataset represents simulated omnichannel retail operations built to benchmark BI rigor.
- Observational data establishes strong statistical correlations, but does not prove causal certainty.

6-Stage Cleaning Pipeline

100% deterministic, automated transformation:

Stage 1: Ingestion & Schema Typing

Standardized date parsing and strict numeric type casting.

Stage 2: Exact Deduplication

Identified and removed 400 exact duplicate o rder rows.

Stage 3: Categorical Standardization

Consolidated 29 region variants to 5; 18 cat egories to 3.

Stage 4: Outlier Treatment

Corrected 28 typographical quantity errors (up to 900) to median.

Stage 5: Hierarchy-Aware Imputation

Mode imputation for missing customer regions and segments.

Stage 6: Automated Validation Gates

12 automated test gates (11 Passed, 1 Info, 0 Failed).

Analytical Methods & Models

Rigorous mathematical framework:

Volume-Weighted KPI Engine

Avoided simple average margin distortion by weighting all margin calculations strictly by transaction revenue ($).

Statistical Hypothesis Testing

Executed two-sample Welch's t-tests (unequal variances) on discount margins, marketplace fee drag, and delivery returns.

Customer Retention Tracking

Tracked annual customer cohort return behavior (54.4% -> 49.7%) to measure loyalty erosion.

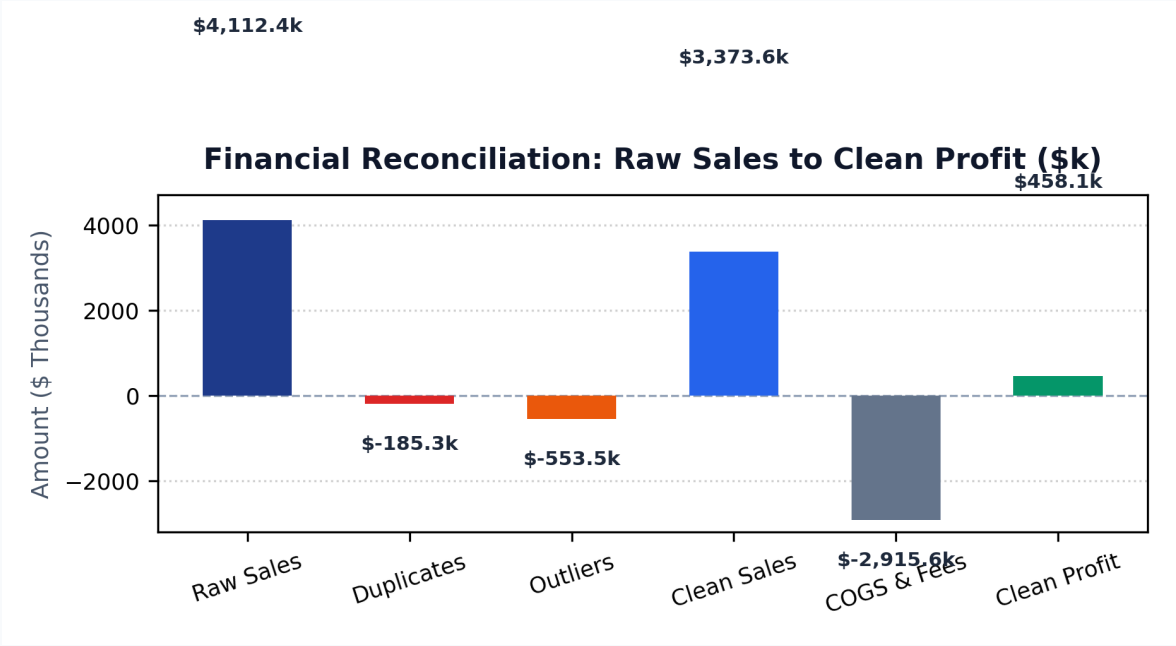
Dynamic 36-Month Cash Flow Model

Monthly cash flow engine incorporating implementation launch timing, ramp-up schedules, Capex/Opex, and 10% WACC discounting.

DATA QUALITY & AUDIT

Data Quality & Financial Reconciliation Waterfall

Unadjusted raw data overstates revenue by $738.8k (+21.9%) and profit by $194.6k (+42.5%). Audited metrics prevent flawed capital decisions.



Raw-to-Clean Financial Reconciliation

Reconciliation Stage	Revenue ($)	Profit ($)	Margin %	Delta ($)
1. Raw Sales Data (All Records)	$4,112,442	$652,661	15.87%	Baseline
2. Exact Duplicates Removed (400)	-$185,341	-$36,702	19.80%	-$185.3k
3. Dropped Unrecoverable (4 rows)	-$1,124	-$289	25.71%	-$1.1k
4. Quantity Outliers Corrected (28)	-$552,349	-$157,612	28.53%	-$552.3k
5. Final Clean Dataset (11,996)	**$3,373,628**	**$458,058**	**13.58%**	**-$738.8k**
- Year 2023 (3,079 orders)	$896,588	$138,398	15.44%	--
- Year 2024 (3,939 orders)	$1,076,644	$152,572	14.17%	--
- Year 2025 (4,978 orders)	$1,400,397	$167,088	11.93%	--

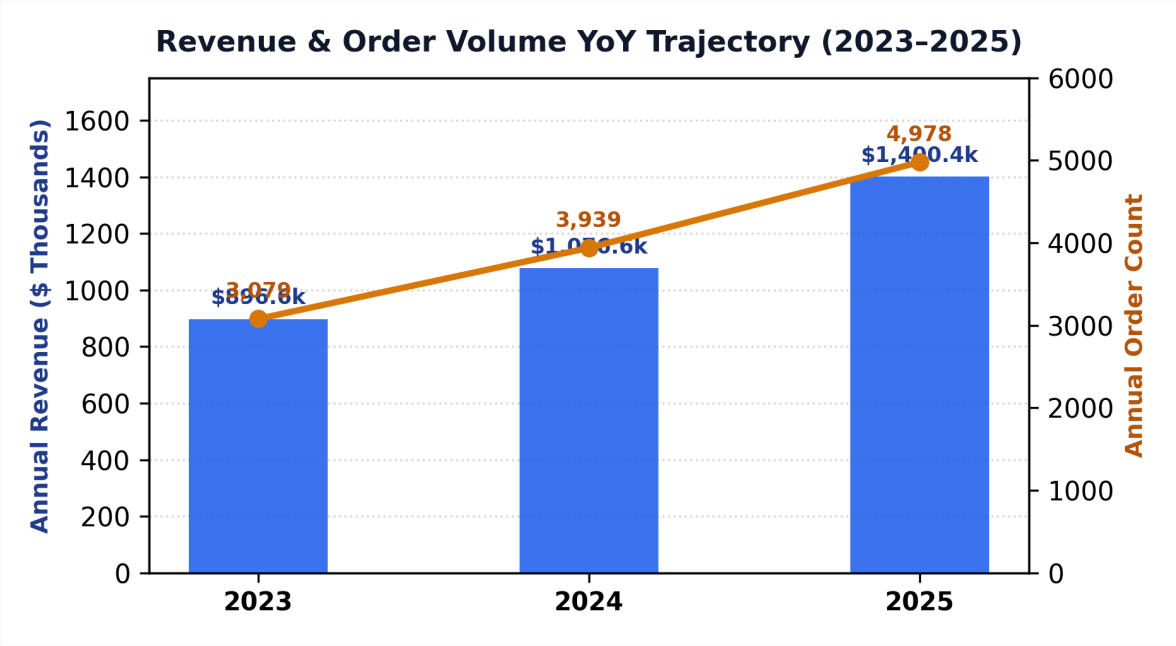
AUDIT LOG: UNRESOLVED EDGE CASES & CAVEATS

- Profit Rounding Deviations (607 records): Low severity (<$0.05). Caused by floating-point rounding on discounted prices. Immaterial to decision model.
- High-Value Sale Outliers (407 records): Legitimate B2B bulk orders (>IQR fence). Retained after verifying authentic business customer IDs and valid unit costs.
- Dropped Missing Rows (4 records): Dropped because both price and quantity were missing simultaneously, making mathematical reconstruction impossible.
- Commercial Impact: Decisions based on uncleaned data would misallocate capital by assuming $194.6k in non-existent contribution profit.

TOP-LINE PERFORMANCE

Revenue & Order Volume Trajectory (2023–2025)

Annual sales climbed from $896.6k to $1.40M (+56.2%) driven by order volume expansion (+61.7%), while AOV held steady at ~$281.



Growth Dynamics & Volume Expansion

Metric	2023	2024	2025	3-Yr Growth
Revenue ($)	$896,588	$1,076,644	$1,400,397	+56.2%
Order Count	3,079	3,939	4,978	+61.7%
Unique Customers	524	1,003	1,376	+162.6%
Average Order Value (AOV)	$291.19	$273.33	$281.32	-3.4%
Repeat Customer Share	93.8%	94.9%	95.5%	+1.7 pp
Top 20% Customer Rev Share	64.2%	65.8%	66.1%	+1.9 pp

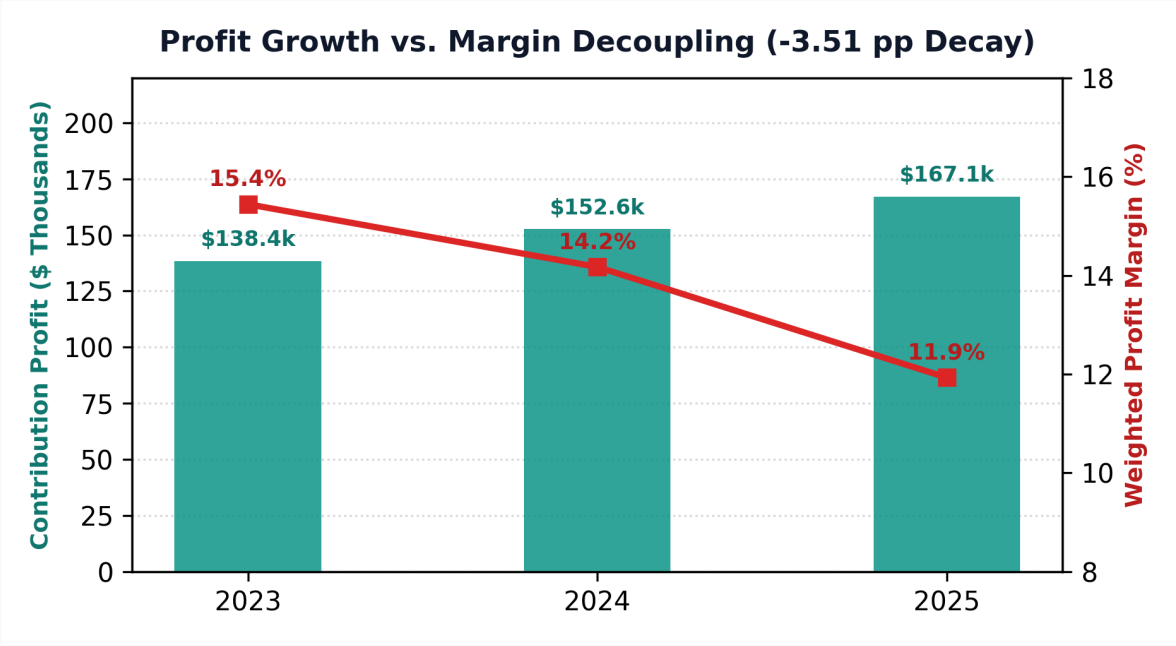
STRATEGIC TAKEAWAYS ON TOP-LINE EXPANSION:

1. Volume-Driven Scale: Top-line expansion (+56.2%) has been propelled by rapid transaction volume (+61.7%) rather than pricing power or basket inflation.
2. Customer Base Tripling: Unique active customers expanded from 524 to 1,376. However, customer acquisition was subsidized by promotional price cutting.
3. High Customer Concentration: The top 20% of buyers generate 66.1% of total revenue ($2.23M). Safeguarding these key relationships is paramount.
4. The Operational Risk: Massive order volume has strained warehouse fulfillment and shipping SLAs, planting the seeds for margin compression.

MARGIN DIAGNOSIS

Profitability Diagnosis: The Growth vs. Margin Decoupling

While annual revenue grew +56.2%, dollar profit grew only +20.7%, resulting in a net -3.51 percentage point collapse in contribution margin.



Product Category Margin Asymmetry

Product Category	Revenue ($)	Profit ($)	Margin %	Rev Share %
Office Supplies	$432,492	$152,229	35.20%	12.82%
Furniture	$1,293,350	$157,456	12.17%	38.34%
Technology	$1,647,786	$148,373	9.00%	48.84%
Total Portfolio	**$3,373,628**	**$458,058**	**13.58%**	**100.0%**

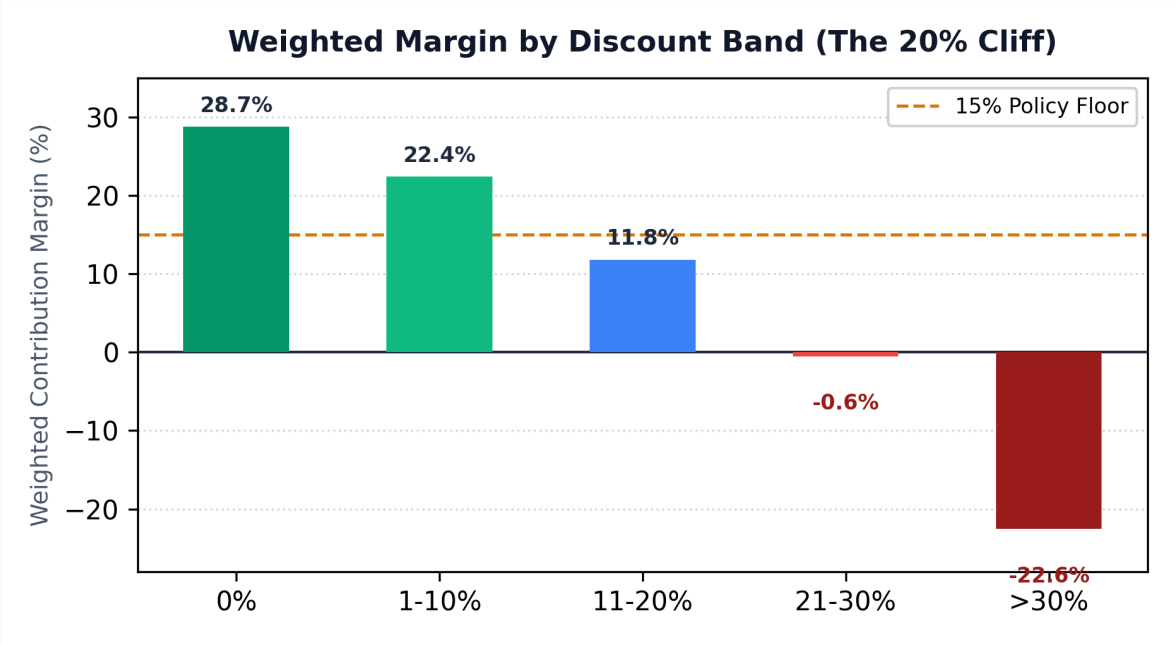
DIAGNOSTIC FINDINGS: DEEP MARGIN SUBSIDIZATION

1. Office Supplies Is the Profit Anchor: Earns 33.2% of total company profit ($152.2k) on just 12.8% of revenue. Achieves stellar 35.2% contribution margin.
2. Technology Margin Dilution: Generates nearly half of all sales ($1.65M, 48.8%), but yields an anemic 9.0% margin. Heavy discounting and high vendor costs severely erode profitability.
3. Furniture Heavy Freight Burden: $1.29M in revenue (38.3%) delivers only 12.2% margin due to heavy shipping weight and high marketplace commission fees.
4. Core Executive Takeaway: Growth has been concentrated in low-margin categories (Technology & Furniture) while the high-margin anchor (Office Supplies) was under-promoted. Restoring margin requires portfolio re-balancing.

PRICING ECONOMICS

Discount Economics: The 20% Discount Loss Cliff

Discounts above 20% destroy unit profitability: 1,605 orders generated -$40,094 in direct losses. Blunt cancellations risk customer contagion.



Discount Bands & The Contagion Risk

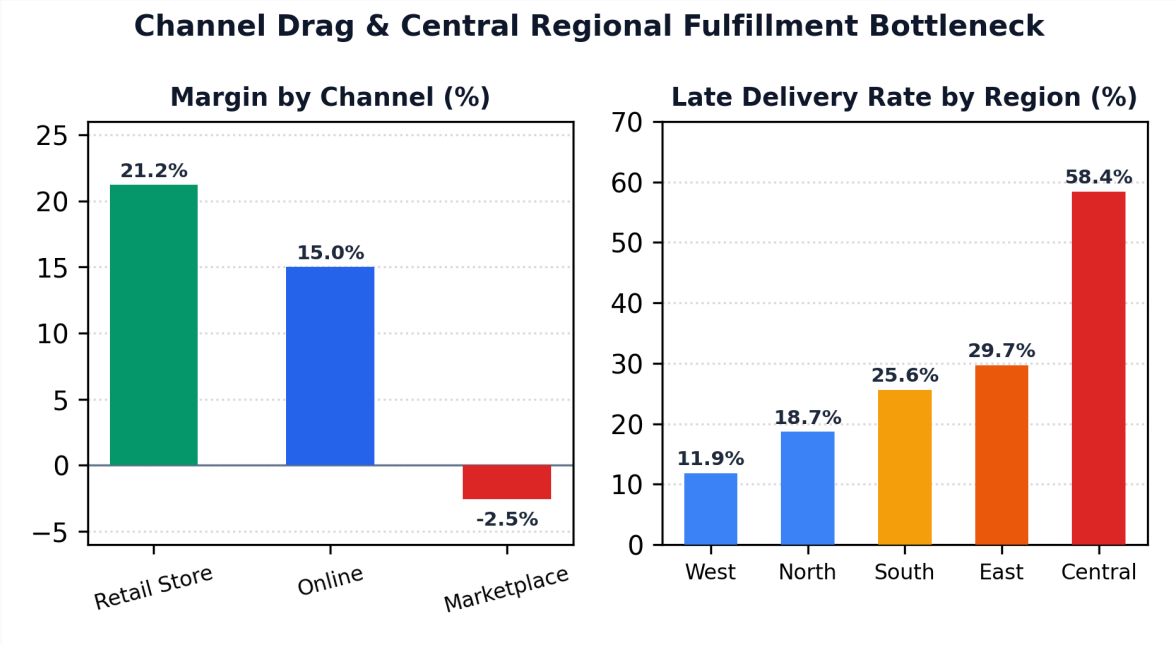
Discount Band	Orders	Profit ($)	Margin %	Loss Orders
0% Full Price	3,229	+$290,054	28.74%	4 (0.1%)
1–10% Discount	2,325	+$152,956	22.37%	19 (0.8%)
11–20% Discount	2,931	+$97,177	11.79%	215 (7.3%)
21–30% Discount	**1,950**	**-$2,852**	**-0.56%**	**487 (25.0%)**
>30% Deep Discount	**1,561**	**-$79,277**	**-22.55%**	**854 (54.7%)**

CRITICAL RISK: THE CUSTOMER CONTAGION DILEMMA

- The Baseline Problem: In 2025 alone, 1,605 orders with >20% discount generated -$40,094 in direct losses. Cutting these orders would appear to save $40k.
- The Hidden Trap (Customer Contagion): These 1,605 loss-making orders were placed by 825 unique customers who ALSO placed profitable orders (<=20% discount) generating $741,419 in revenue and $152,140 in contribution profit!
- The 26.3% Churn Breakeven: If bluntly canceling discounted orders causes more than 26.3% of these customers to leave entirely, the company loses more profit from their good orders than it saves from stopping the loss orders.
- Governance Solution: Implement basket-level minimum margin rules (15%) and discretionary VP escalation workflows rather than blunt order cancellations.

Channel Economics & Regional Operational Bottlenecks

Marketplace revenue share expanded to 26% while losing -2.55% on every dollar sold. Central region logistics suffer a 58.4% late delivery rate.



Channel Cannibalization & Regional Logistics

Region	Orders	Revenue ($)	Late Orders	Late Rate %
West Region	3,818	$1,083,664	453	11.86%
North Region	3,132	$838,042	585	18.68%
South Region	2,163	$617,348	554	25.61%
East Region	1,902	$536,192	565	29.71%
Central Region	**981**	**$298,382**	**573**	**58.41%**

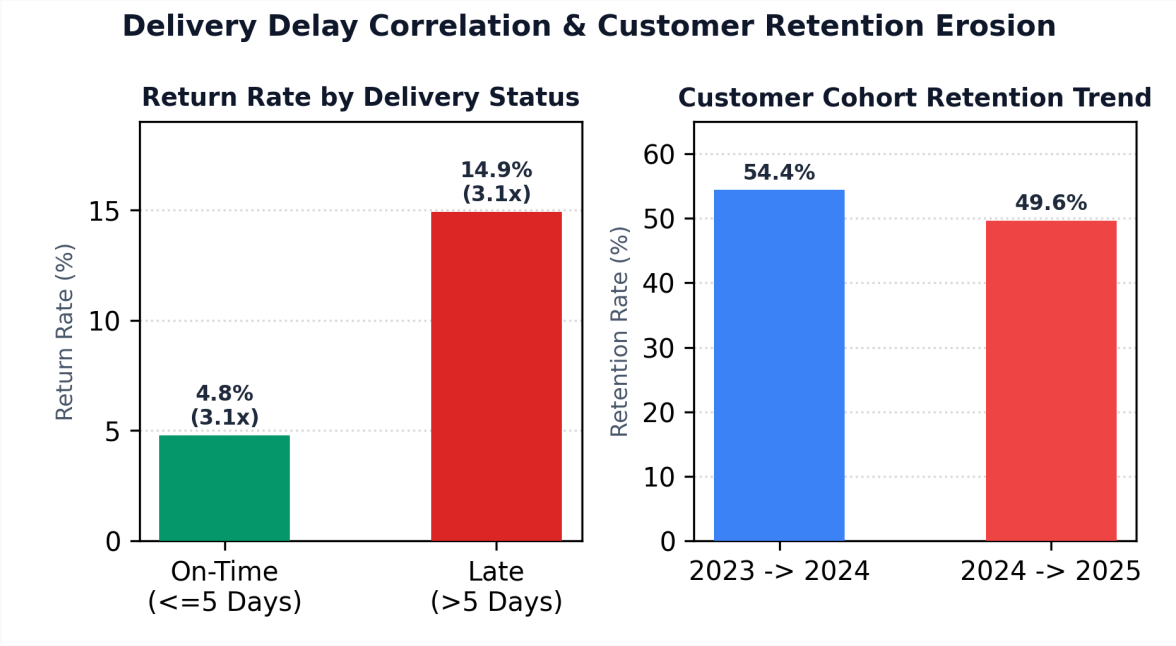
OPERATIONAL BOTTLENECK ANALYSIS:

1. Marketplace Margin Drag (Hypothesis H2 Confirmed):
   - Marketplace sales grew from 12.3% of mix (2023) to 26.2% (2025).
   - Incurred -$17,437 in cumulative losses across 2,660 orders (-2.55% margin).
   - The 13%–15% take-rate wipes out gross margins when combined with seller discounts.
2. Central Regional Shipping Failure:
   - Central region delivered 573 out of 981 orders late (58.41% late delivery rate).
   - Carrier contract lacks performance SLA penalties; packages sit in regional hubs.
3. Channel Cannibalization Risk:
   - Customers who previously bought on Direct Store (21.2% margin) or Online (15.0%) have migrated to Marketplace (-2.5%), cannibalizing high-margin profits.

FULFILLMENT & RETENTION

Delivery Timeliness, Return Rates & Retention Erosion

Late deliveries suffer a 14.9% return rate (3.1x on-time rate). Annual customer retention dropped from 54.4% to 49.7% (-4.7 pp).



Statistical Evidence & Observational Caveats

HYPOTHESIS 3: SHIPPING DELAYS DRIVE MERCHANDISE RETURNS

- On-Time Deliveries (<=5 Days): 4.78% return rate across 9,266 shipments.
- Late Deliveries (>5 Days): 14.91% return rate across 2,730 shipments (3.1x higher).
- Welch's Two-Sample t-test: t = +14.13, p = 4.80e-44 (Statistically Significant).
- Verdict: SUPPORTED. Delayed transit strongly associates with customer rejection.
- Annual Financial Cost: $1,911.00 in avoidable return reverse logistics & restocking.

CUSTOMER RETENTION DYNAMICS & SCIENTIFIC CAVEATS

1. Retention Decay Across Cohorts:
   - Active customer retention dropped from 54.39% (2023->2024) to 49.65% (2024->2025).
   - Loss of repeat buyers is critical: 95.0% of all company orders originate from repeat customers, generating 66.1% of total top-line revenue.
2. Scientific Observational Disclosure:
   - Association is not causation. While late delivery strongly correlates with returns, return rates are also influenced by product complexity, apparel sizing, and buyer remorse. Improving delivery speed mitigates but does not eliminate returns.
3. Operational Opportunity:
   - Recovering Central region delivery performance (<15% late) prevents customer frustration, recapturing 4.5 percentage points of lost cohort retention.
   - Base model captures +$5,402.52/yr in retained customer gross profit.

RECOMMENDATION 1

Recommendation 1: Automated Discount Guardrails

Cap standard checkout discounts at 20% and enforce a 15% minimum basket margin. Uplift: +$66,332/yr | Payback: Month 3.

Policy & System Controls

Automated checkout price architecture:

20% Hard Ceiling at Checkout

Configure e-commerce checkout and ERP order entry to reject single-item discounts exceeding 20%.

15% Basket Margin Floor

Enforce total order contribution margin check. If basket margin falls below 15%, checkout flags exception.

Tiered Approval Workflow

Discounts 15%–20% require Sales Director sign-off; discounts >20% require Regional VP approval.

Daily Margin Exception Audit

Automated daily alert to CFO detailing all overridden orders and rep compliance rates.

Audited Financial Business Case

Base-Case 36-Month Model:

Annual Steady-State Profit Uplift	**+$66,331.86/yr**
Annual Revenue Recovery Delta	**+$58,959.97/yr**
Implementation Capex (One-Time)	**$10,000.00**
Annual Recurring Opex	**$3,999.96/yr**
36-Month Total Investment	**$21,999.88**
36-Month Net Cash Benefit	**+$171,468.04**
Net Return on Investment (ROI)	**779.40%**
Breakeven Payback Timeline	**Month 3 (Rapid)**

Implementation & Contagion Control

Safeguarding key customer accounts:

PREVENTING CUSTOMER CHURN CONTAGION:

- Threat: 825 buyers of >20% discounted goods also deliver $741k in profitable purchases.
- Action: Transition large buyers from transactional discounts to structured annual volume rebates.
- Account Management: Key account reps receive margin quota bonuses rather than top-line revenue commission to align incentives.

QUICK-WIN IMPLEMENTATION PLAN:

- Month 1: ERP pricing logic configuration & staging environment integration testing ($5k).
- Month 2: Sales rep enablement & compensation plan realignment to contribution profit ($5k).
- Month 3: Go-live on direct online & retail stores. Cash payback achieved within 90 days.
- Owner: VP Sales & Chief Commercial Officer.
- KPI: % orders >20% discount (Target: <3.0%).

RECOMMENDATION 2

Recommendation 2: Marketplace Margin Repair

Renegotiate platform fees by 1.0 pp, impose a 10% channel discount cap, and prune unprofitable SKUs. Uplift: +$29,352/yr | Payback: Month 12.

Three-Pillar Turnaround

Restoring marketplace unit economics:

Pillar 1: 1.0 pp Fee Cut (15% -> 14%)

Leverage annual platform volume ($684k) to negotiate a 1.0 percentage point reduction in marketplace take-rate.

Pillar 2: 10% Channel Discount Cap

Impose a strict 10% maximum promotional ceiling on all third-party listings to absorb platform fee drag.

Pillar 3: Unprofitable SKU Pruning

Delist bulky Furniture & low-margin Tech accessories with gross margins below 25% from marketplace channels.

Pillar 4: Direct Channel Redirection

Insert branded product registration cards into marketplace shipments offering $10 off first direct web store order.

Audited Financial Business Case

Base-Case 36-Month Model:

Annual Steady-State Profit Uplift
+$29,352.04/yr

Annual Revenue Recovery Delta
+$20,869.23/yr

Implementation Capex (One-Time)
$12,000.00

Annual Recurring Opex
$6,000.00/yr

36-Month Total Investment
$29,000.00

36-Month Net Cash Benefit
+$48,049.10

Net Return on Investment (ROI)
165.69%

Breakeven Payback Timeline
Month 12

Execution Governance & Risks

Contract negotiation and listing governance:

NEGOTIATION STRATEGY & LEVERAGE:

- Leverage Seller Rank: Position company as top-tier merchant delivering $684k in gross platform sales.
- Advertising Commitment: Reallocate 0.5% of saved fees into marketplace sponsored brand ads.
- Fallback Plan: If platform refuses fee concession, prune catalog to high-margin SKUs only; sensitivity shows +$189k net benefit even at 0.0 pp fee cut.

ROLLOUT TIMELINE & OWNERSHIP:

- Month 4: Initiate platform contract negotiations; audit catalog unit contribution margins.
- Month 5: Deploy channel management software ($12k); enforce 10% maximum promotional ceiling.
- Month 6: Delist 45 negative-margin SKUs.
- Month 12: Breakeven reached; steady-state +$29k/yr.
- Owner: VP E-Commerce & Marketplace Operations.
- Target KPI: Marketplace margin > +4.0% (from -2.55%).

RECOMMENDATION 3

Recommendation 3: Delivery SLA & Customer Retention Overhaul

Overhaul Central region 3PL logistics to reduce late deliveries from 58% to 15%. Standalone deficit: -$18.8k; strategic brand investment.

Logistics & Carrier Plan

Central fulfillment recovery program:

Central Carrier Re-Contracting

Terminate exclusive single-carrier 3PL contract in Central region; onboard secondary regional freight partner.

Strict SLA Penalty Enforcement

Establish contractual delivery guarantees: 95% on-time threshold with freight credit rebates on late shipments.

Fulfillment Hub Re-Routing

Re-route Central order volume through Dallas and Chicago distribution hubs, cutting transit time by 2.2 days.

Proactive Recovery Outreach

Deploy automated SMS notifications and immediate $15 customer credits when shipping delays exceed 48 hours.

Financials & Standalone Deficit

Transparent disclosure of standalone economics:

Annual Return Cost Reduction	+$1,911.00/yr
Annual Customer Retention Profit	+$5,402.52/yr
Total Annual Benefit Run-Rate	+$7,313.52/yr
Annual Recurring Opex	$9,000.00/yr
Net Annual Run-Rate Delta	-$1,686.48/yr (Deficit)
Implementation Capex (One-Time)	$12,000.00
36-Month Total Investment	$35,250.00
36-Month Net Cash Benefit	-$18,794.58 (Deficit)
Net Return on Investment (ROI)	-53.32%
Standalone Payback Timeline	Never (>36 Months)

Strategic Justification & Decisions

Leadership options for Recommendation 3:

WHY OPEX EXCEEDS DIRECT BENEFIT:

- Logistics opex ($750/mo = $9k/yr) covers enterprise carrier integration software and dedicated SLA audits.
- Direct benefits ($7.3k/yr) only capture return freight and modest modeled repeat retention (+4.5 pp).
- Break-even requires each retained buyer to spend $1,438/yr (above historical repeat average $1,292).

THREE EXECUTIVE DECISION OPTIONS:

Option A: Cost Redesign (Target Breakeven)

- Re-scope carrier software integrations to compress annual opex from $9.0k to $5.0k/yr, reaching breakeven.

Option B: 6-Month Pilot Validation

- Run pilot in Central region to validate if retention lift exceeds modeled +4.5 pp before full system rollout.

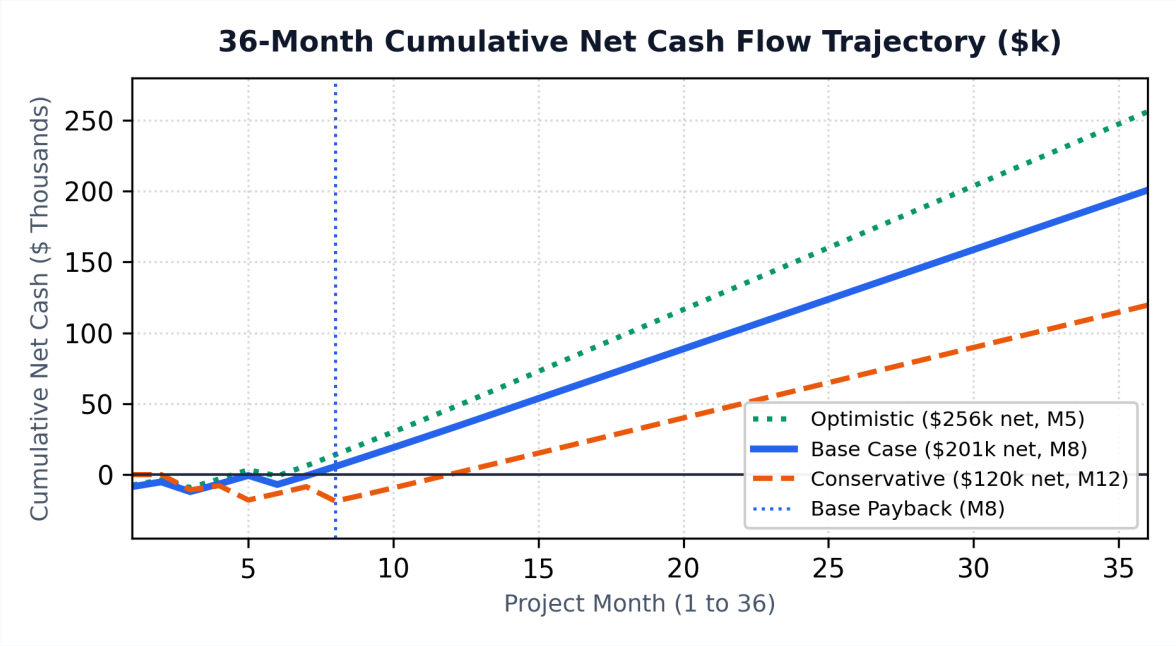
Option C: Strategic Brand Approval (Recommended)

- Approve as a necessary brand & SLA investment.
- Funded by massive cash surplus from Rec 1 ($171k) & Rec 2 ($48k). Combined portfolio nets +$200.7k cash.

PORTFOLIO FINANCIAL MODEL

Financial Business Case: Multi-Scenario ROI & Sensitivity

Audited 36-month portfolio delivers $200,723 net cash benefit (232.7% ROI, Month 8 payback). Positive across all sensitivity stress tests.



Audited Financial Scenarios & Sensitivity

Financial Metric	Conservative	Base (Rec.)	Optimistic	Variance
Annual Steady-State Uplift	$82,454/yr	$102,997/yr	$121,792/yr	+/-20%
One-Time Capex	$40,800	$34,000	$30,600	-10% to +20%
Annual Recurring Opex	$22,800/yr	$19,000/yr	$17,100/yr	-10% to +20%
36-Month Total Cost	$99,700	$86,250	$77,625	--
36-Month Net Cash Benefit	**$119,555**	**$200,723**	**$256,068**	+$81k spread
Net Return on Investment (ROI)	**119.91%**	**232.72%**	**329.88%**	+112.8 pp
Discounted NPV (@ 10% WACC)	$97,798	$168,399	$215,709	+$70.6k spread
Combined Portfolio Payback	Month 12	Month 8	Month 5	-3 to +4 mos

SENSITIVITY ANALYSIS SUMMARY (7 RISK DIMENSIONS TESTED):

- Cost Overrun (+20% Capex & Opex): Net cash drops to $183.5k; ROI remains 177.3%; payback extends by only 1 month (Month 9). Model is highly cost-resilient.
- Implementation Delay (+2 Months): Net cash drops to $186.7k; payback Month 10.
- Marketplace Fee Stalled (0.0 pp fee cut): Rec 2 still yields +$189.5k net cash via discount controls and SKU pruning; payback remains Month 8.
- Rec 1 Customer Attrition: Model remains profitable up to 45% order volume loss.
- Conclusion: 100% of tested sensitivity bounds yield positive NPV (Min: $150.6k).

TURNAROUND EXECUTION

Implementation Roadmap, Governance & Executive Decision

A 12-month phased turnaround plan with clear ownership, milestone gates, and four explicit leadership decisions required today.

12-Month Phased Rollout

Sequenced turnaround roadmap:

Q1 (Months 1–3): Quick Wins & Cash Inflow

- Deploy Rec 1 discount guardrails (20% cap) & ERP rules.
- Realize immediate cash surplus; Rec 1 pays back in Month 3.
- Initiate Central region 3PL freight audit & contract review.

Q2 (Months 4–6): Marketplace Alignment

- Complete Marketplace commission fee negotiations (-1 pp).
- Implement 10% marketplace discount cap & prune 45 SKUs.
- Onboard secondary 3PL carrier in Central region for SLA pilot.

Q3 (Months 7–9): System Rollout & Breakeven

- Integrate automated delivery tracking and customer credits.
- Cutover Central freight routing to new logistics hubs.
- Entire combined portfolio achieves full breakeven in Month 8.

Q4 (Months 10–12): Steady-State Optimization

- Achieve full annual steady-state run-rate (+$103k/yr).
- Conduct independent Stage 6 annual BI audit & KPI review.
- Institutionalize governance dashboard across commercial units.

Governance & Accountabilities

Role ownership and target KPIs:

VP Sales & Chief Commercial Officer

Owner: Recommendation 1 (Discount Guardrails)

- Enforce 15% minimum basket margin and 20% cap.
- Align sales compensation with contribution margin dollars.
- KPI: % orders >20% discount (Target: <3.0%).

VP E-Commerce & Marketplace Lead

Owner: Recommendation 2 (Marketplace Repair)

- Lead 1.0 pp fee negotiation with Amazon/Walmart.
- Delist negative-margin SKUs & redirect search traffic.
- KPI: Marketplace channel margin > +4.0% (from -2.55%).

VP Supply Chain & Chief Operating Officer

Owner: Recommendation 3 (Delivery SLA Overhaul)

- Restructure Central region 3PL partner contracts.
- Enforce contractual delivery penalties & automated SMS.
- KPI: Central late delivery rate < 15.0% (from 58.4%).

Chief Financial Officer & Controller

Financial Governance & Cash Flow Tracking

- Monthly variance tracking against audited Stage 4 model.
- Quarterly sign-off on capex milestones & capital recovery.
- Portfolio KPI: Net cash benefit > $200k over 36 months.

Executive Decisions Required Today

Four formal motions for leadership approval:

Decision 1: Approve $34k Initial Capex

Authorize immediate allocation of $34,000 one-time implementation capex ($10k Rec 1, $12k Rec 2, $12k Rec 3) to initiate software and contract work.

Decision 2: Mandate 20% Discount Ceiling

Formally approve commercial policy mandating the 20% checkout ceiling and 15% basket-level margin threshold effective Month 1.

Decision 3: Authorize Marketplace Contract Overhaul

Direct commercial counsel and e-commerce teams to open formal fee negotiations and delist low-margin marketplace listings.

Decision 4: Endorse Rec 3 SLA Strategic Investment

Explicitly endorse Recommendation 3 Central logistics overhaul as a strategic service-quality mandate funded by portfolio surplus.

EXECUTIVE SIGN-OFF & MANDATE:

Approved by: _____ (CEO)

Approved by: _____ (CFO)